A3-PS003-TC520

Federated Traffic-Flow Prediction
with Communication-Cost Evaluation

TEAM
Chauraha Sync
Aryan Jha · aryanjha.ad26@bmsce.ac.in
Apeksha Benneshiroor · apekshabenneshiroor.ec26@bmsce.ac.in

FACULTY MENTOR
Dr. Monika Puttaramaiah
monika.mel@bmsce.ac.in
BMS College of Engineering · Bengaluru, India

GitHub: github.com/jhaaryan7580-dev/A3-PS003-TC520

LinkedIn: linkedin.com/in/aryan-jha-785a29343

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Problem Description & Abstract

What PS-003 asks

- Can locations with different traffic patterns learn one shared model fairly?
- How much accuracy changes when training stays distributed?
- How do IID-style and non-IID client groups differ in error and convergence?
- When is federated communication cheaper than a one-time training-flow upload?

Scale: 17,856 × 170 × 3 tensor; 5 clients × 34 sensors; 3 seeds; 30 FedAvg rounds.

Abstract overview

We build a reproducible CPU-only simulator on public PEMS08 traffic-flow data, used as a stand-in for heterogeneous locations.

Five simulated clients are compared across persistence, daily-cycle, centralised, local-only, IID FedAvg and non-IID FedAvg conditions.

Measured 3-seed MAE: centralised 13.938; IID FedAvg 13.642; local-only 14.011; non-IID FedAvg 14.017.

Communication: 615,600 bytes per 30-round FedAvg run versus an 8,499,320-byte training-flow upload reference.

Measurement study; not a new algorithm, Bengaluru dataset, or privacy guarantee.

Metrics are test-set means over seeds 0, 1 and 2 unless explicitly marked as audit or derived arithmetic.

Background & Methodology

Dataset and motivation
PEMS08: **17,856 × 170 × 3**; 62 days; 5-minute steps; channel 0 flow target.
Derived from California Caltrans PeMS. It is a public stand-in, not Bengaluru traffic.

Federated Averaging
Five clients train locally; the server averages weights by local sample count.
Each round: **2,052 B down + 2,052 B up per client** = **20,520 B**.

Quality audit
NaN = **0**; Inf = **0**; zero-flow count = **10,560** (**0.348%**).
Max zero run = **400**; mean zero-run length = **2.005**; lag-288 autocorrelation = **0.839**.

IID-style versus non-IID
IID-style random control W1 distance = **28.542**.
Quiet-to-busy non-IID W1 distance = **127.867**, or **4.48×** higher.
Primary zero rule: complete-case windows; forward-fill is sensitivity-only.

Literature & RFC Review

Algorithmic core
[1] McMahan et al., 2017
FedAvg defines local training, sample-weighted averaging and communication accounting.
[2] Liu et al., 2020
FedGRU establishes federated traffic forecasting as a valid direction.
Project translation: sample-weighted FedAvg; **513 parameters**; **20,520 B/round**.

Data & benchmarks
[3] Guo et al., 2019
ASTGCN provides the PEMS08 benchmark and provenance context.
[4] Caltrans PeMS
Official source context for California detector data.
Project translation: 12 lag values + 2 cycle features; **3 seeds × 4 learned conditions = 12 result files**.

Graph optimisation & fairness
[5] Banik et al., 2025
Motivates transparent comparison against graph-aware aggregation.
[6] Panda et al., 2025
Motivates client-level error analysis.
Project translation: client MAE gap = **3.323** IID versus **10.836** non-IID.

RFC context: no RFC defines federated traffic forecasting; we borrow reproducibility discipline, not standards authority.

Proposed Solution

Pipeline and model
1. Audit: hash the tensor; count **10,560** zeros; verify **0** NaN/Inf.
2. Split: chronological **70/15/15**; windows = **2,122,790 / 453,220 / 453,390**.
3. Features: 12 previous valid flows + sine/cosine of index position t mod 288.
4. Model: 14 → 32 → 1 MLP; **513 parameters**; **2,052 float32 bytes**.
5. Clients: five groups × 34 sensors; IID-style control and quiet-to-busy non-IID bands.

Conditions and evaluation
Six comparisons: persistence, daily-cycle, centralised, local-only, IID FedAvg and non-IID FedAvg.
FedAvg: all 5 clients participate; one local epoch; 30 rounds; sample-weighted averaging.
Communication: 5 × 2 directions × 2,052 = **20,520 B/round**; 30 rounds = **615,600 B**.
Reference: 12,499 × 170 × 4 = **8,499,320 training-flow bytes**; 13.81× the 30-round FedAvg total.
Metrics: MAE, RMSE, relative error, client gap, rounds and bytes; means ± SD over seeds 0/1/2.
Stand-in data; simulated clients; no privacy guarantee or live deployment claim.

Optimization Proposed

Completed test results: mean ± sample SD over seeds 0, 1 and 2; 170 sensors; one local epoch.

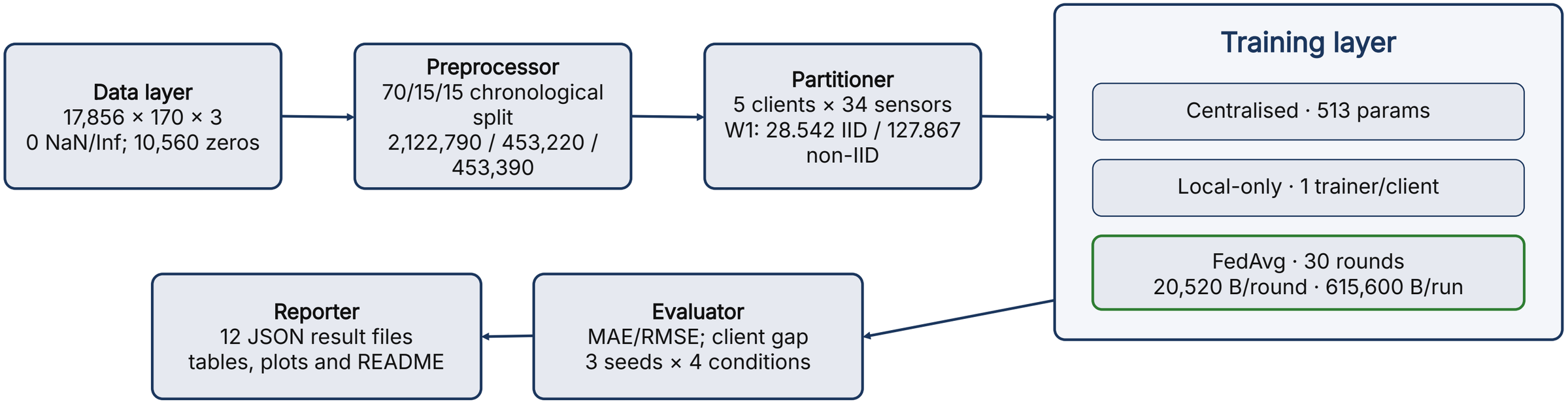
Condition	Test MAE	Test RMSE	Bytes / run
Persistence	16.037	24.547	—
Time-of-day	89.734	117.428	—
Centralised	13.938 ± 0.037	21.224 ± 0.028	0
Local-only	14.011 ± 0.009	20.543 ± 0.021	0
FedAvg IID-style	13.642 ± 0.017	21.158 ± 0.020	615,600
FedAvg non-IID	14.017 ± 0.268	21.412 ± 0.186	615,600

Observed: IID FedAvg is 2.12% lower MAE than centralised in the 3-seed matrix and 14.93% below persistence. Non-IID FedAvg is 0.57% above centralised and has a client MAE gap of **10.836** versus **3.323** for IID FedAvg.
These are measured outcomes, not claims of generalisation beyond PEMS08.

Solution Architecture & Design

Every component exposes a count, shape, or measured metric.

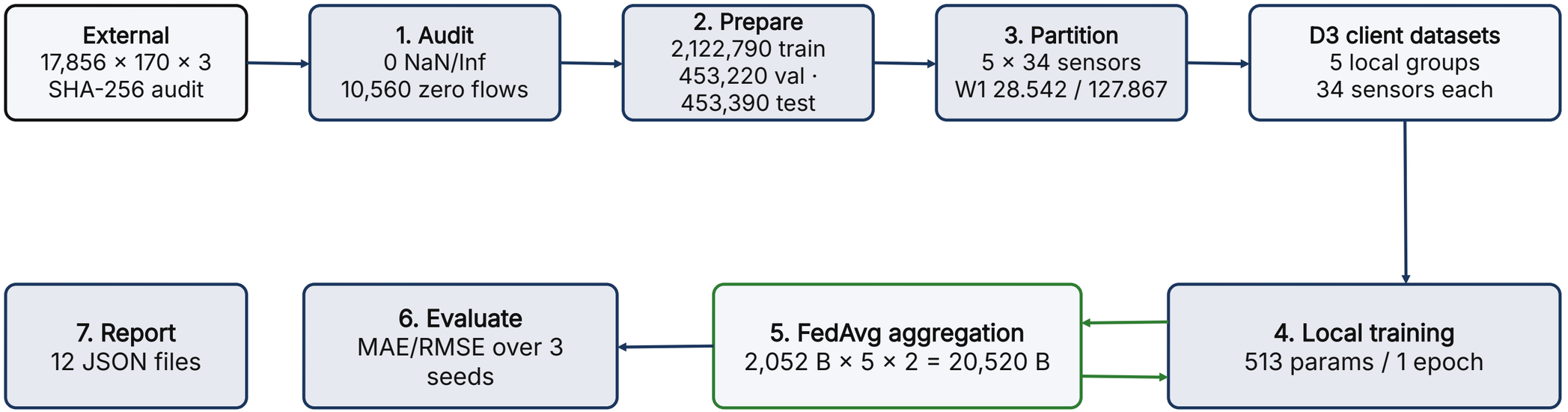
3 seeds × 4 learned conditions = 12 result files; all learned conditions use 170 sensors.



Measured output: IID FedAvg MAE 13.642 ± 0.017; non-IID FedAvg MAE 14.017 ± 0.268.

Data Flow Diagram

Counts are attached to each data or model transfer.



30 rounds = 615,600 FedAvg bytes/run.
Training-flow upload lower bound = 8,499,320 bytes.

Figure: Data Flow Diagram (DFD); raw windows remain in client datasets.

Deployment Timeline

Every milestone has a measurable exit condition.

Date	Milestone	Exit check
Oct 7–13	Level 1 repository, ownership transfer, presentation and team video.	Steps 1–7 ready by Oct 13; stated deadline Oct 14.
Oct 10–14	Exams. No feature development; freeze code and keep restart note.	Only scripted/resumable work when safe.
Oct 16	Level 2 announcement and transition to next stage.	Team checks organizer instructions and scope.
Oct 16–31	Mentoring sessions and review checkpoints.	Mentor reviews zero rule, upload reference and core metrics.
Oct 17–21	Reproduce audit: 0 NaN/Inf, 10,560 zero flows; run baselines and first FedAvg.	Audit JSON and baseline MAE/RMSE reproduced.
Oct 22–26	Run 3 seeds × 4 learned conditions = 12 result files.	MAE/RMSE, client gaps and 615,600 B/run logged.
Oct 27–29	Demo from saved results, report PDF, README and pseudocode.	Demo link live; figures generated by scripts.
Oct 30–31	Fresh-clone reproduction test and final submission package.	Steps 8–10 complete by Oct 31.
Oct 31–Nov 4	Evaluators may connect with teams during the review window.	Connection possible, not guaranteed; submission already due.

Cut order: zero-policy sensitivity first, then E=5 local epochs, then the 5% scarcity level. Never cut 3 seeds, baselines, chronological split or byte counter.

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[7] AIORI-3 Hackathon. “PS-003 planning brief and proposed structure template.” github.com/aiori-hackathon/A3-PSXXX-TCXXX

Thank You for Your Attention!

Point of contact
Aryan Jha · aryanjha.ad26@bmsce.ac.in
Apeksha Benneshiroor · apekshabenneshiroor.ec26@bmsce.ac.in
Dr. Monika Puttaramaiah · monika.mel@bmsce.ac.in

GitHub
github.com/jhaaryan7580-dev/A3-PS003-TC520

LinkedIn
linkedin.com/in/aryan-jha-785a29343

Scan to open the repository
A3-PS003-TC520

Questions and feedback from mentors are welcome.